



DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001-2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

DEPARTMENT	ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING				
Semester: VII	Course Title: Data Security & Privacy Lab			Course Code: 22AI73	
PROGRAM QUESTION	Phishing Website Detection using Python: Use a dataset of phishing and legitimate websites. Train a simple ML model using scikit-learn to classify URLs based on features like length, presence of @, https, etc.				
STUDENT NAMES	Udayaraj		USN	1DS22AI057	
LAB MARKS	Program Execution (10)	Source Code (10)	Result (5)	Innovation/ Extra Features (5)	Total
TOOLS USED	Cursor				
MAPCOURSE OUTCOME	CO1				
MAP PROGRAM OUTCOME	PO2, PO3, PO4, PO5				
MAP PROGRAM SPECIFIC OUTCOME	PSO1, PSO2				
UML / SEQUENCE DIGRAM	<pre>sequenceDiagram actor User participant System participant Model1 as Model participant Model2 as Model User->>System: Submit URL for checking System->>System: Extract features (length, @, https, etc.) System->>Model1: Predict label using features Model1-->>Model2: Label (Phishing/Legitimate) Model2-->>System: System-->>User: Return result (Phishing/Legitimate)</pre>				
COURSE COORDINATOR	Dr ARUNA M G & Prof ENSTEIH SILVIA				

Code for Program 5:

5) Phishing Website Detection using Python: Use a dataset of phishing and legitimate websites. Train a simple ML model using scikit-learn to classify URLs based on features like length, presence of @, https, etc

```
import streamlit as st, pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

def extract_features(url):
    return {"url_length": len(url), "has_at": int("@" in url), "has_https": int(url.startswith("https")),
            "num_dots": url.count("."), "has_hyphen": int("-" in url)}

@st.cache_data
def load_data():
    data = {"url": ["http://example.com", "https://secure-login.com", "http://phishing.com@evil.com",
                  "http://paypal.login.verify.com", "https://google.com", "http://192.168.0.1/login",
                  "https://microsoft.com", "http://update-your-bank.com"],
            "label": [0, 0, 1, 1, 0, 1, 0, 1]}
    df = pd.DataFrame(data)
    features = df["url"].apply(lambda x: pd.Series(extract_features(x)))
    return pd.concat([df, features], axis=1)

@st.cache_resource
def train_model(df):
    X, y = df[["url_length", "has_at", "has_https", "num_dots", "has_hyphen"]], df["label"]
    model = LogisticRegression().fit(*train_test_split(X, y, test_size=0.3, random_state=42)[:2])
    acc = accuracy_score(y, model.predict(X))
    return model, acc

st.title("🕵️ Phishing Website Detector")
df = load_data()
model, acc = train_model(df)
st.sidebar.info(f"Model accuracy: {acc*100:.2f}%")

url_input = st.text_input("Enter URL:")
if st.button("Check URL") and url_input.strip():
    X_new = pd.DataFrame([extract_features(url_input)])
    pred = model.predict(X_new)[0]
    st.error("🚨 Phishing Site!") if pred else st.success("✅ Legitimate Site!")
    with st.expander("🔍 Features"): st.json(X_new.to_dict(orient='records')[0])
```

RESULT:

